EMERALD COAST

GLISTENING GEM OF THE SOUTH

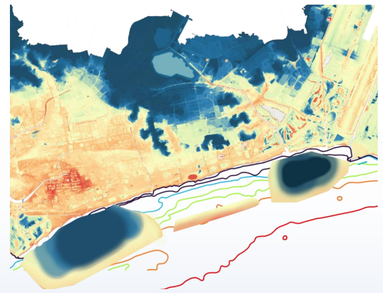
Employing cutting edge coastal protection strategies, it practices prudential planning against the rising sea levels along Singapore's southeast shoreline, while achieving regenerative urban design that revitalizes the country's ecological diversity. Consisting 3 islands, our proposal is human-centered as it is nature-centered. Stringing Changi City and Founders Memorial, Emerald Coast concentrates diverse experience of the people, weaving a tapestry of leisure, travel, heritage, business, luxury, education, retail, housing and greenery into the landscape. The name is a reflection of the spirit of our design – combining hybrid engineering and human creativity, Emerald Coast gleams in the light of nature, shines in the vitality of human activity.

- Commercial & Residential
- Residential
- Park
- Beach Area
- Reserve Site
- Water Body
- Open Space
- Commercial
- Business



Long Island Reclamation will be divided into West, Middle and East Region. West Region caters for residential living with highlights of recreational and tourism. Middle Region shows a beach area with commercial activities. East Region will be developed as a new business area. A reservoir is located in the middle, becoming Singapore's largest reservoir.

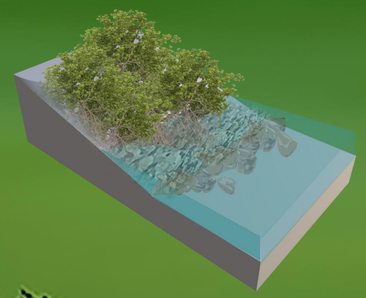
DEM (mSHD)	Water Depth (mSHD)
0	0 - 2
2	2 - 5
4	5 - 15
6	15 - 20
8	20 - 50
10	
>15	

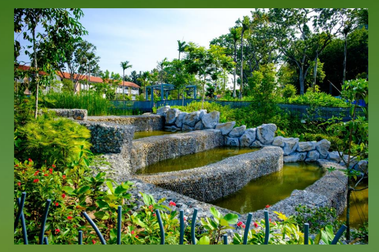


East Coast Park has a low-lying terrain level of 1.0 – 3.0 mSHD. With Long Island Reclaimed land building at a minimum platform level of 4.0mSHD, the reservoir water level is maintained at the current sea level.

MANGROVE FOREST TRAIL

Mangroves act as a natural defence against coastal erosion and sea level rise by dissipating wave energy and reducing the impact on the shore. It improves water quality and provides wildlife habitats. Mangrove systems have high adaptability with the natural ability to adapt to sea level rise. A trail will be built along the mangrove trees lead to the Founders Memorial in the north and the Coral Paradise on the East, promoting ecotourism.





WETLAND BIO-GARDEN

Wetland bio-gardens located near the west end of East Coast Park act as carbon sinks and natural buffers against flooding. Promote biodiversity conservation by providing habitats for various plant and animal species. Suitable for outdoor activities and environmental education.

CORAL PARADISE



The coral reefs will serve crucial functions such as reducing erosion, stabilizing the beach, and providing habitat for marine species.



The artificial tidal pool improves biodiversity by reducing desiccation and heat stress. It offers education, conservation, research, and recreation benefits.

The coral museum and marine gallery will offer interactive exhibits and a hands-on learning experience for visitors, focusing on coral reef ecosystems, and marine exploration, conservation.

XBLOC

Xbloc is an interlocking concrete armour unit that reduces wave energy and erosion while promoting biodiversity by creating natural habitat within the hollow design. It's made from recycled materials with a low carbon footprint, which makes it durable and reduces the need for maintenance.



The blocks can be stacked and arranged in different configurations to accommodate various site conditions, such as wave heights, water depths, and seabed slopes, which make it a flexible option to encounter 4m total sea water rise and is easy to adapt even when it rise to 5m.



LONG ISLAND BARRAGE

Barrage is important in flood control and storing water. It includes a gate structure with an air pumping and emission system, plus a water expulsion feature. The pumping station is used to remove excess water during heavy rainfall and pump seawater during drought periods.

The installation of a crest gate, and pumping station, is effective for managing stormwater runoff, regulating water levels and runoff, and mitigating flooding during heavy rainfall. The design is flexible and can be modified to accommodate a higher sea level. For instance, increasing the height of the gate and pumping capacity.

PASIR TENGAH

At Pasir Tengah, a long stretch of perched beach will be nourished at a location with steep shoreface. It includes a submerged breakwater to hold foreshore nourishment and facilitate wave attenuation and erosion protection. A substitute to the existing East Coast Park's beaches for leisure activities.



RESERVOIR

The reservoir has three main functions: water supply, flood control, and lifestyle attraction. It stores stormwater for drinking water and prevents seawater from flooding low-lying areas by maintaining a constant water level of 1.0-2.0mSHD at the existing East Coast area.

WATER PLAZA

Water plazas feature effective solutions for stormwater management and improving the quality of stormwater runoff. Incorporating Urban Water Buffers with sustainable design and green infrastructure into the city's stormwater management system.

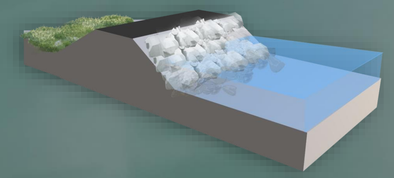
AMPHITHEATRE

Barrage-cum-amphitheatre serves as a new convergence of Singapore's cultural and recreational expression. It incorporates Urban Water Buffers such as featuring a green roof to manage stormwater runoff effectively, reducing the urban heat island effect.

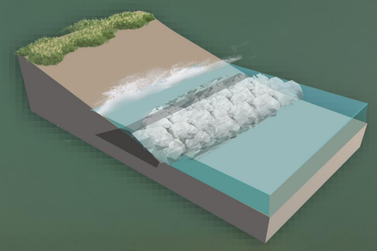
Xbloc groynes safeguard against erosion caused by negative net longshore transport gradients and rising sea levels. Designed to withstand the forces of waves and currents.

COASTAL STRUCTURES

Xbloc dike serves to prevent flooding of low-lying hinterlands caused by rising sea levels. It is a watertight structure made up of large concrete blocks that interlock to form a barrier, reducing wave energy and erosion.



Xbloc breakwaters reduce wave impact on the shoreline and withstand the forces of waves and currents. Protecting property and reducing the risk of flooding.



Xbloc revetments are more resistant to wave action and erosion than traditional revetments. The rough element of concrete helps to reduce wave overtopping.

